

Lightweight Rover Chassis Truss

Design and test a lightweight truss chassis that supports a specified payload.

Intermediate	2-3 class periods	Engineering Challenge
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Engineering Context

A planetary rover chassis must protect instruments while minimizing launch mass. Your team will create a small truss structure and compare strength-to-mass performance.

What You Will Practice

- Identify tension and compression members
- Use triangulation effectively
- Measure structural performance
- Calculate strength-to-mass ratio

What You Need

- Craft sticks, paper strips, or approved structural material
- Adhesive
- Mass scale
- Test weights
- Ruler
- Engineering notebook

Student Instructions

- 1 Define the required chassis span and payload location.
- 2 Sketch two truss patterns and label expected tension/compression members.
- 3 Estimate material quantity and mass for each concept.
- 4 Select and build one chassis using consistent joints.
- 5 Measure final mass and inspect alignment.
- 6 Load the chassis in approved increments while observing deformation.
- 7 Record maximum successful load and failure location.
- 8 Calculate load-to-mass ratio and recommend one redesign.

Engineering Requirements

- Span the required distance without intermediate support
- Support the assigned base payload for 30 seconds
- Use a clearly identifiable truss pattern
- Record mass before testing
- No repair during an official load trial

Constraints & Safety

- Use only approved adhesives and cutting tools
- Wear eye protection during destructive testing if required
- Keep hands and faces away from the load path
- Do not exceed instructor load limits

What You Submit

- Two concept sketches
- Member-force predictions
- Structure mass
- Load-test table
- Failure analysis
- Redesign recommendation

Reflection Questions

- 1 Which members appeared to be in compression?
- 2 Where did the first visible deformation occur?
- 3 How did joint quality affect performance?
- 4 What change would improve strength without adding much mass?

Engineering Tip Compression members often fail by buckling before the material itself crushes.	Common Mistake Adding material everywhere increases mass but may not strengthen the critical load path.
Stretch Goal Rebuild one member or joint and demonstrate a 15% improvement in load-to-mass ratio.	Career Connection Aerospace structures engineers design trusses, frames, and support structures to carry loads with minimal mass.